

Danish Shakil

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Professional Summary

Mechanical engineering professional with hands-on CAD experience delivering client-facing design work, including a completed manufacturing-process project for British American Tobacco (BAT) and a product enclosure design for a US-based trailer manufacturer. Skilled in 3D modeling, mechanical simulation, and interpreting technical specifications, with a track record of managing design revisions and meeting delivery requirements in remote, multidisciplinary teams.

Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Topi, Pakistan

Bachelor of Science in Mechanical Engineering

Expected Graduation: 2027

CGPA: 3.29/4.00

Technical Skills

- **CAD & Modeling Software:** SolidWorks (CSWA Certified), Blender
- **Engineering & Analysis Tools:** ANSYS, COMSOL Multiphysics, MATLAB, Simulink, Dynacam, Linkages
- **Robotics & Embedded Systems:** ROS 2, RViz2, Proteus, Arduino IDE
- **Core Competencies:** Machine Design (Shigley's standards), Technical Drawing Interpretation, GD&T Fundamentals, Topology Optimization, Geometric Inverse Kinematics
- **Manufacturing:** 3D Printing, Rapid Prototyping, Additive Manufacturing

Professional & Client CAD Experience

Mechanical Design Intern — British American Tobacco (BAT)

Jun 2025 – Jul 2025

Automated Tobacco Reclaimer System

- Designed and mapped an automated tobacco reclaimer system as part of a manufacturing internship, aimed at optimizing primary production operations.
- Produced technical design reports and detailed cross-departmental process flow diagrams for engineering and operations stakeholders.

CAD Design Contractor — Bigfoot Trailers (US-based client)

Aug 2025

Yeti 24 Product Enclosure Design

- Delivered the complete 3D CAD model for a Yeti-branded product enclosure for a US-based trailer manufacturer, working remotely to client specifications.
- Ensured designs strictly adhered to client requirements, optimizing structure for manufacturability and aesthetic appeal.

Academic & Competition CAD Projects

RoboCup Small Size League Robot

Dec 2024 – May 2025

- Developed 3D CAD assemblies for an autonomous soccer robot equipped with Maxon motors.
- Designed and modeled a specialized mechanical dribbler and kicker mechanism in SolidWorks, optimizing packaging constraints and internal clearances.
- Managed iterative design revisions for rapid prototyping and executed 3D printing of complex structural components.

Heavyweight Combat Robot (60 kg Vertical Spinner)

Jul 2024 – May 2025

- Executed structural calculations, material selection, and force analysis to ensure design integrity under heavy impact.
- Developed the full 3D CAD assembly, ensuring all internal mechanisms met dimensional and weight constraints.
- Transitioned CAD models into physical builds through hands-on fabrication, machining, and rapid prototyping.

Aerospace Bracket Topology Optimization

Dec 2024 – May 2025

- Executed a topology optimization study on a Ti-6Al-4V titanium alloy aerospace bracket.
- Used finite element analysis (FEA) to refine CAD geometry, minimizing weight while meeting structural deflection and stress limits.

Certifications

- Dassault Systèmes: Certified SOLIDWORKS Associate (CSWA)
- SOLIDWORKS Electrical Design
- Additive Manufacturing